THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 6 Solution
2026/03/04 and 2026/03/06

1. At a party N men throw their hats into the center of a room. The hats are mixed up and each man randomly selects one. Let M be the number of men who select their own hat. Find $\mathbb{E}[M]$ and $\mathrm{Var}(M)$.

Solution:

Let X_i be the indicator random variable for the event that the i-th man gets his own hat, i.e.

$$X_i = \begin{cases} 1 & \text{if the } i\text{-th man gets his own hat,} \\ 0 & \text{otherwise.} \end{cases}$$

Then $M = X_1 + \cdots + X_N$. By linearity of expectation,

$$\mathbb{E}[M] = \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_N].$$

For any $i = 1, \ldots, N$, the i-th man gets each of the N hats with equal probability. We have

$$\mathbb{E}[X_i] = \mathbb{P}(X_i = 1) = \frac{1}{N}.$$

Thus $\mathbb{E}[M] = N \cdot \frac{1}{N} = 1$. For the variance, we have

$$\mathrm{Var}(M) = \mathrm{Var}\left(\sum_{i=1}^{N} X_i\right) = \sum_{i=1}^{N} \mathrm{Var}(X_i) + 2 \cdot \sum_{i<j} \mathrm{Cov}(X_i, X_j).$$

Note that X_i^2 has the same distribution as X_i, so

$$\mathrm{Var}(X_i) = \mathbb{E}[X_i^2] - \mathbb{E}[X_i]^2 = \mathbb{E}[X_i] - \mathbb{E}[X_i]^2 = \frac{1}{N} - \frac{1}{N^2}.$$

On the other hand, for all $i \neq j$ we have

$$\begin{aligned} \mathbb{E}[X_i X_j] &= \mathbb{P}(X_i = 1, X_j = 1) \\ &= \mathbb{P}(\text{both the } i\text{-th and the } j\text{-th man gets the correct hats}) \\ &= \frac{1}{N(N-1)}. \end{aligned}$$

Thus $\mathrm{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j] = \dfrac{1}{N^2(N-1)}$, and

$$\mathrm{Var}(M) = N\left(\frac{1}{N} - \frac{1}{N^2}\right) + 2 \cdot \frac{N(N-1)}{2} \frac{1}{N^2(N-1)} = 1.$$

2. Suppose $X_1, X_2, \ldots, X_n$ are independent and identically distributed (i.i.d.) random variables with mean μ and variance σ^2. Let the sample mean be defined by

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

(a) Find $\mathbb{E}[\bar{X}]$.

(b) Find $\mathrm{Var}(\bar{X})$.

(c) Find $\mathrm{Cov}(\bar{X}, X_i - \bar{X})$ for $i = 1, 2, \ldots, n$.

Solution:

(a) By linearity of expectation,

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \mu.$$

(b) By the independence of $X_1, \ldots, X_n$,

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{\sigma^2}{n}.$$

(c) For any $i = 1, \cdots, n$, we have

$$\begin{aligned}\mathrm{Cov}(\bar{X}, X_i - \bar{X}) &= \mathbb{E}[\bar{X}(X_i - \bar{X})] - \mathbb{E}[\bar{X}]\cdot\mathbb{E}[X_i - \bar{X}] \\ &= \left(\mathbb{E}[\bar{X}X_i] - \mathbb{E}[\bar{X}]\cdot\mathbb{E}[X_i]\right) - \left(\mathbb{E}[\bar{X}^2] - \mathbb{E}[\bar{X}]^2\right) \\ &= \mathrm{Cov}(\bar{X}, X_i) - \mathrm{Var}(\bar{X}).\end{aligned}$$

Similarly, one can show that

$$\mathrm{Cov}(\bar{X}, X_i) = \mathrm{Cov}\left(\frac{1}{n}\sum_{j=1}^{n} X_j, X_i\right) = \frac{1}{n}\sum_{j=1}^{n}\mathrm{Cov}(X_j, X_i) = \frac{1}{n}\mathrm{Var}(X_i) = \frac{\sigma^2}{n}.$$

Therefore,

$$\mathrm{Cov}(\bar{X}, X_i - \bar{X}) = \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0.$$

Remark: The above arguments can be generalized to show that covariance is bilinear, i.e. for any random variables X, Y, Z over the same sample space (not necessarily independent) and $a, b \in \mathbb{R}$, we have

$$\mathrm{Cov}(aX + bY, Z) = a\mathrm{Cov}(X, Z) + b\mathrm{Cov}(Y, Z),$$

$$\mathrm{Cov}(X, aY + bZ) = a\mathrm{Cov}(X, Y) + b\mathrm{Cov}(X, Z).$$

Meanwhile, part (c) provides another example that zero covariance does not imply independence. (Why isn't $\bar{X}$ and $X_i - \bar{X}$ independent in general? Try to construct a simple example.)

3. (a) Let $X_1, X_2, \ldots, X_n$ be i.i.d. Poisson random variables with parameters $\lambda_1, \lambda_2, \ldots, \lambda_n$, respectively. Prove that the sum $Y = X_1 + X_2 + \cdots + X_n$ is a Poisson random variable with parameter $\lambda = \lambda_1 + \lambda_2 + \cdots + \lambda_n$.

(b) Let $X_1, X_2, \ldots, X_n$ be i.i.d. exponential random variables with parameter λ. Prove that the sum $Y = X_1 + X_2 + \cdots + X_n$ is a gamma random variable with parameters n and λ.

Solution:

This question can be solved in an easier way by using moment generating functions. However, to illustrate the computation of probability mass/density function of sum of random variables, we provide a more elementary argument here.

(a) We prove by induction on n. The base case $n = 1$ is trivial. Assume that the sum of n i.i.d. Poisson random variables is a Poisson random variable with parameter $\lambda = \lambda_1 + \cdots + \lambda_n$. Let $Z = X_1 + \cdots + X_n$ and $Y = X_1 + \cdots + X_{n+1}$. We thus have $Z \sim \text{Poi}(\lambda)$ from the induction hypothesis, $Y = Z + X_{n+1}$, and

$$\begin{aligned} p_Y(x) &= \sum_k p_Z(k) p_{X_{n+1}}(x-k) = \sum_{k=0}^{x} \frac{e^{-\lambda}\lambda^k}{k!} \frac{e^{-\lambda_{n+1}}\lambda_{n+1}^{x-k}}{(x-k)!} \\ &= e^{-(\lambda+\lambda_{n+1})} \sum_{k=0}^{x} \frac{\lambda^k \lambda_{n+1}^{x-k}}{k!(x-k)!} = \frac{e^{-(\lambda+\lambda_{n+1})}}{x!} \sum_{k=0}^{x} \frac{x!}{k!(x-k)!} \lambda^k \lambda_{n+1}^{x-k} \\ &= \frac{e^{-(\lambda+\lambda_{n+1})}}{x!} \sum_{k=0}^{x} \binom{x}{k} \lambda^k \lambda_{n+1}^{x-k} \\ &= \frac{e^{-(\lambda+\lambda_{n+1})}}{x!} (\lambda + \lambda_{n+1})^x. \end{aligned}$$

Note that the right hand side is exactly the probability mass function of $\text{Poi}(\lambda + \lambda_{n+1})$, which completes the induction.

(b) We prove by induction on n. The base case $n = 1$ is trivial from comparing the probability density function of $\text{Exp}(\lambda)$ and $\text{Gamma}(1, \lambda)$. Assume that the sum of n i.i.d. exponential random variables is a gamma random variable with parameters n and λ. Let $Z = X_1 + \cdots + X_n$ and $Y = X_1 + \cdots + X_{n+1}$. We thus have $Z \sim \text{Gamma}(n, \lambda)$ from the induction hypothesis, $Y = Z + X_{n+1}$, and

$$\begin{aligned} f_Y(x) &= \int_y f_Z(y) f_{X_{n+1}}(x-y) dy \\ &= \int_0^x \frac{\lambda^n}{(n-1)!} y^{n-1} e^{-\lambda y} \cdot \lambda e^{-\lambda(x-y)} dy \\ &= \frac{\lambda^{n+1}}{(n-1)!} e^{-\lambda x} \int_0^x y^{n-1} dy \\ &= \frac{\lambda^{n+1}}{n!} x^n e^{-\lambda x}, \end{aligned}$$

which is exactly the probability density function of $\text{Gamma}(n+1, \lambda)$. Thus $Y \sim \text{Gamma}(n+1, \lambda)$, and the induction is complete.